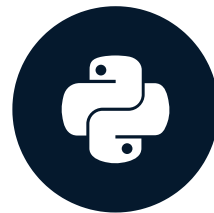


Python dictionaries

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What is a dictionary?

- Similar to a MATLAB structure array
- Collection of key:value pairs
- Ordering is not guaranteed

Retrieving data from a dictionary

```
type(definitions)
```

```
dict
```

```
print(definitions['armadillo'])
```

```
A nocturnal burrowing mammal with long ears, a tubular snout, and a long  
extensible tongue, feeding on ants and termites. Armadillos are native to  
Africa and have no close relatives.
```

Creating dictionaries

```
# Create a dictionary  
dog = {'name': 'Toby', 'breed': 'Basset Hound'}
```

Adding data to a dictionary

```
# Add more key:value pairs
dog['weight (lbs)'] = 552.3
dog['birthdate'] = "2016-06-26"
print(dog)
```

```
{'name': 'Toby',
 'breed': 'Basset Hound',
 'weight (lbs)': 552.3,
 'birthdate': '2016-06-26'}
```

Updating dictionaries

```
# Update weight value  
dog['weight (lbs)'] = 52.3  
print(dog)
```

```
{'name': 'Toby',  
 'breed': 'Basset Hound',  
 'weight (lbs)': 52.3,  
 'birthdate': '2016-06-26'}
```

Removing data from a dictionary

```
dog.pop('birthdate')  
print(dog)
```

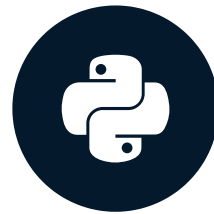
```
{'name': 'Toby',  
 'breed': 'Basset Hound',  
 'weight (lbs)': 52.3}
```


Let's practice!

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Introduction to DataFrames

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DataFrames

- `pandas` package
- Tabular data
- `pandas DataFrame` = MATLAB table
- Store data where each observation has mixed types:
 - floats
 - integers
 - Booleans
 - strings

DataFrames

```

      rank  num_households  frac_dog_owners  frac_cat_owners
state
Alabama      17      1828000          0.441          0.274
Arizona      18      2515000          0.401          0.296
Arkansas       6      1148000          0.479          0.306
California    40     12974000          0.328          0.283
Colorado      13      1986000          0.425          0.323
Connecticut   33      1337000          0.283          0.319
Delaware      25       334000          0.337          0.337
District of Columbia 48       287000          0.131          0.116
Florida       32      7609000          0.357          0.273
Georgia       31      3798000          0.401          0.273
```

.head() method

```
pets.head()
```

	rank	num_households	frac_dog_owners	frac_cat_owners
state				
Alabama	17	1828000	0.441	0.274
Arizona	18	2515000	0.401	0.296
Arkansas	6	1148000	0.479	0.306
California	40	12974000	0.328	0.283
Colorado	13	1986000	0.425	0.323

.columns attribute

```
pets.columns
```

```
Index(['rank', 'num_households', 'frac_dog_owners', 'frac_cat_owners'],  
      dtype='object')
```

.index attribute

```
pets.index
```

```
Index(['Alabama', 'Arizona', 'Arkansas', 'California', 'Colorado',  
      'Connecticut', 'Delaware', 'District of Columbia', 'Florida',  
      'Georgia', 'Idaho', 'Illinois', 'Indiana', 'Iowa', 'Kansas',  
      'Kentucky', 'Louisiana', 'Maine', 'Maryland', 'Massachusetts',  
      'Michigan', 'Minnesota', 'Mississippi', 'Missouri', 'Montana',  
      'Nebraska', 'Nevada', 'New Hampshire', 'New Jersey', 'New Mexico',  
      'New York', 'North Carolina', 'North Dakota', 'Ohio', 'Oklahoma',  
      'Oregon', 'Pennsylvania', 'Rhode Island', 'South Carolina',  
      'South Dakota', 'Tennessee', 'Texas', 'Utah', 'Vermont', 'Virginia',  
      'Washington', 'West Virginia', 'Wisconsin', 'Wyoming'],  
      dtype='object', name='state')
```

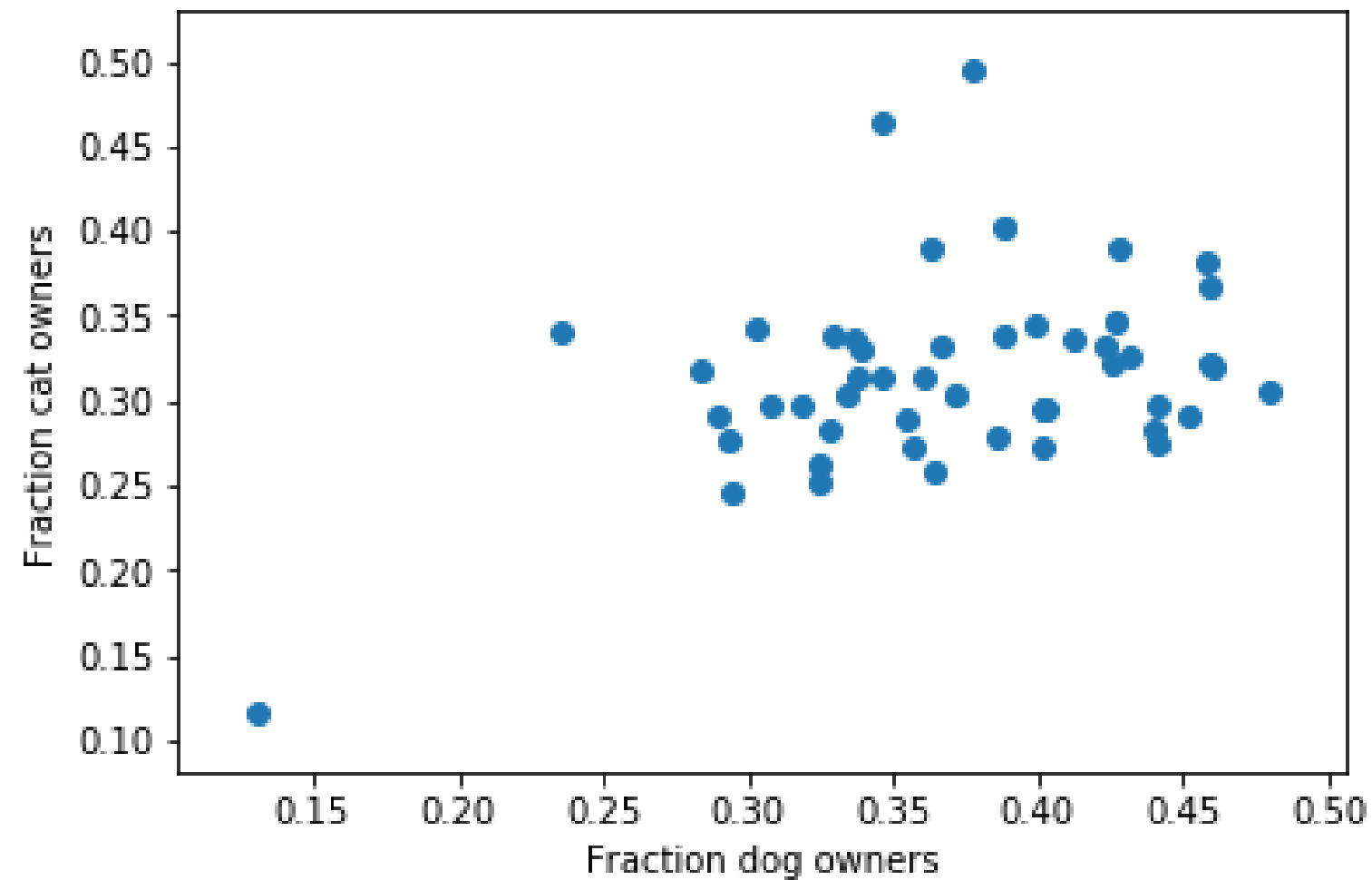
Getting one column out

```
pets['rank']
```

```
state
Alabama      17
Arizona      18
Arkansas       6
California    40
...
Virginia     37
Washington    5
West Virginia  7
Wisconsin     21
Wyoming       9
Name: rank, dtype: int64
```


NumPy & Matplotlib compatible

```
plt.scatter(pets['frac_dog_owners'], pets['frac_cat_owners'])
```

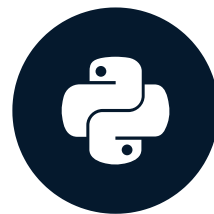


Let's practice!

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Accessing pandas DataFrames

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Selecting multiple dataframe columns

```
pets[['frac_dog_owners', 'rank']]
```

	frac_dog_owners	rank
state		
Alabama	0.441	17
Arizona	0.401	18
Arkansas	0.479	6
California	0.328	40
...		
Virginia	0.354	37
Washington	0.363	5
West Virginia	0.458	7
Wisconsin	0.339	21
Wyoming	0.388	9

Selecting rows

1. pandas indexing with `.loc`
2. Python indexing with `.iloc`

Selecting rows using pandas indexing

```
pets.loc['Massachusetts']
```

```
rank          47.000  
num_households 2618000.000  
frac_dog_owners  0.236  
frac_cat_owners  0.341  
Name: Massachusetts, dtype: float64
```

Selecting rows using Python indexing

```
pets.iloc[17]
```

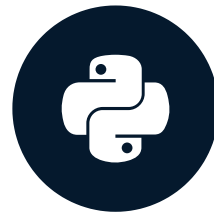
```
rank          4.000  
num_households 548000.000  
frac_dog_owners 0.346  
frac_cat_owners 0.464  
Name: Maine, dtype: float64
```

Let's practice!

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Creating pandas DataFrames

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```
Date, AveragePrice, Total Volume, 4046, 4225, 4770
2015-12-27, 1.33, 64236.62, 1036.74, 54454.85, 48.16
2015-12-20, 1.35, 54876.98, 674.28, 44638.81, 58.33
2015-12-13, 0.93, 118220.22, 794.70, 109149.67, 130.50
2015-12-06, 1.08, 78992.15, 1132.00, 71976.41, 72.58
2015-11-29, 1.28, 51039.60, 941.48, 43838.39, 75.78
```

```
import pandas as pd
avocados = pd.read_csv('avocados.csv')
avocados.head()
```

	Date	AveragePrice	Total Volume	4046	4225	4770
0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16
1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33
2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50
3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58
4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78

```
avocados = pd.read_csv('avocados.csv', index_col=0)
```

From a dictionary of lists

```
import pandas as pd
pd.DataFrame()
forecast_raw = { 'weekday': ['Mon', 'Tues', 'Wed', 'Thurs', 'Fri', 'Sat', 'Sun'],
                  'rain': [True, False, False, False, True, True, False],
                  'temp': [68, 72, 73, 75, 67, 68, 68] }
forecast = pd.DataFrame(forecast_raw)
print(forecast)
```

	rain	temp	weekday
0	True	68	Mon
1	False	72	Tues
2	False	73	Wed
3	False	75	Thurs
4	True	67	Fri
5	True	68	Sat
6	False	68	Sun

From a list of dictionaries

```
forecast_raw = [ {'rain': True, 'temp': 68, 'weekday': 'Mon'},
                  {'rain': False, 'temp': 72, 'weekday': 'Tues'},
                  {'rain': False, 'temp': 73, 'weekday': 'Wed'},
                  {'rain': False, 'temp': 75, 'weekday': 'Thurs'},
                  {'rain': True, 'temp': 67, 'weekday': 'Fri'},
                  {'rain': True, 'temp': 68, 'weekday': 'Sat'},
                  {'rain': False, 'temp': 68, 'weekday': 'Sun'}, ]

import pandas as pd
forecast = pd.DataFrame(forecast_raw)
print(forecast)
```

	rain	temp	weekday
0	True	68	Mon
1	False	72	Tues
2	False	73	Wed
3	False	75	Thurs
4	True	67	Fri
5	True	68	Sat
6	False	68	Sun

Let's practice

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